Induction machine fundamentals

Answers and worked solutions · Day 10 · Mon 2 Nov · pack M7

5. Practice

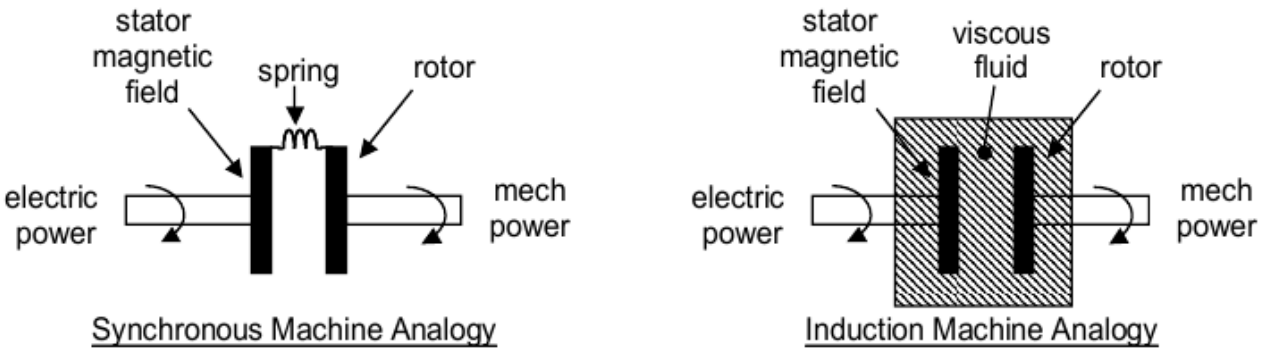
From your course files

PQ 7.1 (practice questions).

a. A **short-circuited three-phase winding**.
b. Any three of: low cost, self-starting, high efficiency, reliable, available in a wide range of sizes.
c. **Squirrel-cage** rotor. The bars are normally **aluminium**.
d. The **slip speed** $n_S - n$.
e. Generator when the rotor speed is **greater** than n_S. Motor when it is **less**.
f. A **transformer** (with a short-circuited secondary).
g. A **resistor**: $R_2'(1-s)/s$.
h. **Starting torque** is the torque at zero speed ($s = 1$). **Pull-out torque** is the maximum torque the motor can produce at any speed.

PQ 7.2 (practice questions).

- **Synchronous machine:** join the disks with a **spring**. Both turn at exactly n_S. The twist of the spring (the rotor angle) sets the torque.
- **Induction machine:** put a **viscous fluid** between the disks. The rotor disk lags the field disk. The torque is proportional to the speed difference (slip speed). Power flows from the faster disk to the slower one.



PQ 7.2 official answer (from the practice questions): a spring between the disks for the synchronous machine, a viscous fluid around the disks for the induction machine

PQ 7.3 (practice questions).

$n_S = 120f/\text{poles}$:

a. $120(50)/4 =$ **1500 rpm**. (b) $120(60)/8 =$ **900 rpm**. (c) $120(50)/2 =$ **3000 rpm**. (d) $120(800)/4 =$ **24 000 rpm**. (e) $120(50)/24 =$ **250 rpm**.

PQ 7.4 (practice questions).

Try $f = 50$ and 60 Hz in poles $= 120f/n_S$ and keep the answer that is an even integer:

a. 1500 rpm: 50 Hz gives 4 (60 Hz gives 4.8). **4 poles, 50 Hz.**
b. 1800 rpm: 60 Hz gives **4 poles, 60 Hz**.
c. 750 rpm: 50 Hz gives **8 poles, 50 Hz** (60 Hz gives 9.6).
d. 1200 rpm: 60 Hz gives **6 poles, 60 Hz** (50 Hz gives 5, which is odd).
e. 360 rpm: 60 Hz gives **20 poles, 60 Hz** (50 Hz gives 16.67).
f. 150 rpm: **40 poles at 50 Hz** or **48 poles at 60 Hz**. Both are valid.

PQ 7.5 (practice questions).

Method: $p = \lfloor 60f/n_R \rfloor$, $n_S = 60f/p$, slip speed $= n_S - n_R$ (and, for PQ 7.8, $s_R = (n_S - n_R)/n_S$, $f_2 = s_R f$). The voltage is irrelevant.

Motor	$60f/n_R$	p	Poles	n_S (rpm)	Slip speed (rpm)	s_R (PQ 7.8)	f_2 (PQ 7.8)
(a) 415 V, 50 Hz, 1440 rpm	2.083	2	**4**	**1500**	**60**	**0.04000**	**2.000 Hz**
(b) 3.3 kV, 60 Hz, 3520 rpm	1.023	1	**2**	**3600**	**80**	**0.02222**	**1.333 Hz**
(c) 135 V, 50 Hz, 680 rpm	4.412	4	**8**	**750**	**70**	**0.09333**	**4.667 Hz**
(d) 230 V, 50 Hz, 950 rpm	3.158	3	**6**	**1000**	**50**	**0.05000**	**2.500 Hz**

All values match the PQ answers.

PQ 7.8 (practice questions).

These are the last two columns of the PQ 7.5 table: (a) 0.04, 2 Hz (b) 0.02222, 1.333 Hz (c) 0.09333, 4.667 Hz (d) 0.05, 2.5 Hz. For example (a): $s_R = 60/1500 = 0.04$ and $f_2 = 0.04 \times 50 = 2$ Hz.

PQ 7.6 (practice questions).

Use the linear torque–speed line through $(n_S, 0)$ and (n_R, τ_R) (Lecture 7.2.2), so $\tau = \tau_R\,(n_S - n)/(n_S - n_R)$.

a. $60(50)/2910 = 1.031$, so $p = 1$, $n_S =$ **3000 rpm**. Rated slip speed $=$ **90 rpm**.

b. At no load τ = 0, so slip speed = 0 and $n =$ **3000 rpm** (ideally).

c. Rated torque occurs at rated speed, **2910 rpm**.

$$\omega_R = 2910 \times \frac{2\pi}{60} = 304.7 \text{ rad/s}, \qquad \tau_R = \frac{7500}{304.7} = \mathbf{24.61 \text{ Nm}}$$

d. $2910/3000 =$ **97 %**. Yes: the speed drops only 3 % from no load to full load, so "almost fixed speed" is fair.

e. Half torque means half the slip speed, 45 rpm. $n = 3000 - 45 =$ **2955 rpm**.

f. $P = \tau\omega = 12.31 \times 2955 \times 2\pi/60 =$ **3.808 kW**, which is **50.77 %** of rated. It is more than half because the speed (2955 rpm) is higher than rated speed. In fact $P/P_R = 0.5 \times 2955/2910$.

Disagreement with the PQ answer: the PQ gives 3.801 kW and 50.68 %. Those numbers are $12.31 \times 2950\,\text{rpm}$, i.e. the answer used 2950 rpm instead of its own 2955 rpm from part (e). With 2955 rpm the correct values are 3.808 kW and 50.77 %. The explanation (higher speed) is the same.

g. Slip speed $= 90 \times 20/24.61 = 73.14$ rpm, so $n = 3000 - 73.14 =$ **2927 rpm**.

h. 3070 rpm is above n_S, so it is a **generator**. Slip speed $= -70$ rpm.

$$\tau = 24.61 \times \frac{-70}{90} = \mathbf{-19.14 \text{ Nm}}$$

The negative sign means the shaft drives the machine.

PQ 7.7 (practice questions).

$60(50)/1470 = 2.04$, so $p = 2$, $n_S = 1500$ rpm and the rated slip speed is 30 rpm.

a. Half torque means half the slip speed (15 rpm): $n = 1500 - 15 =$ **1485 rpm**.

b. A generator at rated torque has a slip speed of −30 rpm: $n = 1500 + 30 =$ **1530 rpm**.

PQ 7.9 (practice questions).

$60(50)/960 = 3.125$, so $p = 3$, $n_S = 1000$ rpm. $R_2' = 1\,\Omega$.

Case	n (rpm)	$s = (1000 - n)/1000$	$f_2 = s \times 50$	R_2'/s
(a) locked rotor	0	**1**	**50 Hz**	**1 Ω**
(b) no load	1000	**0**	**0 Hz**	**∞ (open circuit)**
(c) rated	960	**0.040**	**2 Hz**	**25 Ω**
(d) generator	1050	**−0.050**	**−2.5 Hz**	**−20 Ω**

A negative R_2'/s means the "resistor" delivers power: mechanical power is absorbed and appears as electrical power. All values match the PQ answers.

PQ 7.10 (practice questions).

a. **Greater** than synchronous speed.
b. The difference between synchronous speed and rotor speed, $n_S - n$.
c. In the induction machine the secondary is short-circuited, the effective rotor resistance R_2'/s depends on slip, and the magnetising current is much larger because of the airgap.
d. $n = 0.9n_S$, so $s =$ **0.1** and $f_2 = 0.1 \times 50 =$ **5 Hz**.
e. The **airgap power** P_{AG}.
f. **Stator copper loss** and **(stator) iron loss**.
g. The **rotor copper loss**, sP_{AG}. (The question says "less than"; it means P_{AG} is *greater* than P_M.)
h. The rotor frequency sf is very low. Iron loss rises with frequency (eddy-current loss with f^2), so the rotor iron loss is negligible.
i. R_1 from a DC resistance test on the stator. $R_1 + R_2'$ from the locked-rotor test, so $R_2' = R_{LR} - R_1$ (detail in pack M8).
j. i. Low slip: $\tau \propto s$. (ii) High slip: $\tau \propto 1/s$.
k. Squirrel cage: R_2' is fixed. Wound rotor: R_2' can be increased with external three-phase resistors through slip-rings.
l. Changing R_2' changes the **speed (slip)** at which maximum torque occurs, but not its **value**.
m. **Starting** torque.
n. Ideal $\eta = 1 - s = n/n_S =$ **85 %**.
o. At no load $I_2' \approx 0$, so the input current is almost all magnetising current through X_M. That current lags V_1 by nearly 90°, so the pf is low.
p. About **2**.
q. $R_2'/s = R_2' + R_2'(1 - s)/s$. R_2' is the real rotor-winding resistance: the power in it (sP_{AG}) is heat. $R_2'(1 - s)/s$ is a fictitious resistance: the power in it ($(1 - s)P_{AG}$) is

the electrical power converted to mechanical power.

PQ 7.11 (practice questions).

This is the Type D method in Section 2.

a. $60(50)/1420 = 2.11$, so $p = 2$ and $n_S =$ **1500 rpm**.

b. $\omega_R = 2\pi \times 1420/60 = 148.7$ rad/s, so $\tau_R = 2200/148.7 =$ **14.80 Nm**.

c. i. $P_{IN} = \sqrt{3} V_L I_L \cos\phi = \sqrt{3}(415)(4.9)(0.78) =$ **2.747 kW**. (ii) $P_{IN} = P_{OUT}/\eta = 2200/0.80 =$ **2.750 kW**. They agree.

d. Ideal $\eta = n_R/n_S = 1420/1500 =$ **94.67 %**. The ideal efficiency only counts the rotor copper loss. Stator copper loss, iron loss and friction and windage make the real efficiency (80 %) lower.

Course answers: 1500 rpm, 14.80 Nm, 2.747 kW and 2.750 kW, 94.67 %.

PQ 4.7 (practice questions, only if you skipped M6).

a. $f = 60 \times 900/1200 =$ **45 Hz** and, keeping V/f constant, $V = 110 \times 45/60 =$ **82.5 V**.

b. The slip speed at a given torque is unchanged (100 rpm rated, 50 rpm at half torque): (i) **800 rpm** (ii) **850 rpm**. Full working in pack M6.

New questions

N1. [3 marks]

$n_S = 120(60)/6 = 1200$ rpm. $s = (1200 - 1164)/1200 =$ **0.03**. $f_2 = 0.03 \times 60 =$ **1.8 Hz**.

$R_2'/s = 0.45/0.03 =$ **15 Ω**, which splits into $R_2' =$ **0.45 Ω** (rotor copper loss) and $R_2'(1-s)/s =$ **14.55 Ω** (mechanical power).

N1: R2′/s = 15 ohm split into R2′ = 0.45 ohm and R2′(1-s)/s = 14.55 ohm

N2. [3 marks]

$n_S = 1000$ rpm, $s = 0.03$.

$P_{AG} = P_{\text{rotor Cu}}/s = 400/0.03 =$ **13.33 kW**. $P_M = (1-s)P_{AG} =$ **12.93 kW**.

$\tau = P_{AG}/\omega_S = 13333/104.72 =$ **127.3 Nm**. Check: $P_M/\omega = 12933/101.58 = 127.3$ Nm.

N3. [5 marks]

a. Delta, so $V_1 = V_L = 400\angle 0^\circ$ V. $s = 45/1500 = 0.03$, $R_2'/s = 33.33\,\Omega$.

$$Z = (1.2 + 33.33) + j4 = 34.76\angle 6.607^\circ\ \Omega, \qquad I_1 = I_2' = \frac{400}{34.76\angle 6.607^\circ} = \mathbf{11.51\angle -6.607^\circ\ A}$$

Line current $I_L = \sqrt{3} \times 11.51 =$ **19.93 A**.

b. $P_{AG} = 3(11.51)^2(33.33) =$ **13.24 kW**. $\tau = 13239/157.08 =$ **84.28 Nm**.
$P_{IN} = 3(400)(11.51)\cos 6.607^\circ = 13.72$ kW. $P_M = 0.97 \times 13.24 = 12.84$ kW.
$\eta = 12.84/13.72 =$ **93.63 %**.

Trap: if you use $400/\sqrt{3}$ (star thinking), you get $\tau = 28.09$ Nm, exactly one third.

N4. [10 marks]

a. $60(50)/1440 = 2.08$, so $p = 2$, $n_S = 1500$ rpm, $\omega_S = 157.1$ rad/s. $s = 60/1500 =$ **0.04**. $R_2'/s =$ **15 Ω**. Circuit: $V_1 = 415/\sqrt{3} = 239.6\angle 0^\circ$ V across $500\,\Omega \parallel j40\,\Omega$, then the series branch $0.8\,\Omega + j3\,\Omega + 15\,\Omega$.

N4(a) circuit with values: 239.6 V, 500 ohm parallel j40 ohm, then 0.8 ohm, j1.5 ohm, j1.5 ohm and R2′/s = 15 ohm

b. $Z = 15.8 + j3 = 16.08\angle 10.75^\circ\ \Omega, \qquad I_2' = \dfrac{239.6}{16.08\angle 10.75^\circ} = \mathbf{14.90\angle -10.75^\circ\ A} = 14.64 - j2.779$ A

$$I_M = \frac{239.6}{500} - j\frac{239.6}{40} = 0.4792 - j5.990\ \text{A}$$

$I_1 = 15.12 - j8.769 = \mathbf{17.48\angle -30.12^\circ\ A}, \qquad \text{pf} = \cos 30.12^\circ =$ **0.8650 lagging**

c. $P_{IN} = 3(239.6)(17.48)(0.8650) =$ **10.87 kW**. $P_{AG} = 3(14.90)^2(15) =$ **9.988 kW**.
$\tau = 9988/157.08 =$ **63.59 Nm**. The torque formula gives the same:
$3(239.6)^2(15)/\{157.08[(15.8)^2 + 3^2]\} = 63.59$ Nm.

d. $P_M = (1-s)P_{AG} = 0.96 \times 9988 =$ **9.589 kW**. $\eta = 9.589/10.87 =$ **88.25 %**. Ideal $\eta = 1 - s = 96\,\%$. Losses: stator Cu $3(14.90)^2(0.8) = 532.7$ W, iron $3(239.6)^2/500 = 344.5$ W, rotor Cu $sP_{AG} = 399.5$ W. Then $9589 + 399.5 + 532.7 + 344.5 = 10\,865$ W $= P_{IN}$. The ideal figure counts only the rotor copper loss. The stator copper and iron losses (877 W) take it from 96 % down to 88.25 %.

N5. [5 marks]

a. $\tau \propto V_1^2$ at fixed slip: $\tau = 0.9^2 \times 63.59 =$ **51.51 Nm**.

b. At low slip, $\tau \approx 3V_1^2 s/(\omega_S R_2')$. For constant τ, $s \propto 1/V_1^2$: $s \approx 0.04/0.81 = 0.04938$, so $n \approx 1500(1 - 0.04938) =$ **1426 rpm**.

c. Put $x = R_2'/s$ and $V_1 = 0.9 \times 239.6 = 215.6$ V into the torque formula:

$$3V_1^2 x = \tau\omega_S \left[(R_1 + x)^2 + X^2\right] \;\Rightarrow\; 9988x^2 - 123\,521x + 96\,287 = 0$$

The roots are $x = 11.53\,\Omega$ (the stable, low-slip root) and $x = 0.836\,\Omega$ (beyond pull-out). So $s = 0.6/11.53 =$ **0.05204** and $n =$ **1422 rpm**. Since τ is unchanged, $P_{AG} = \tau\omega_S = 9988$ W is also unchanged. Rotor Cu $= sP_{AG} = 0.05204 \times 9988 =$ **519.7 W** (up from 399.5 W, +30 %), and I_2' rises from 14.90 A to 16.99 A. Comment: a voltage sag on a constant-torque load makes the motor run slower, draw more rotor current and run hotter. The linear estimate is close (1426 vs 1422 rpm) but slightly optimistic, because R_1 and X are not negligible.

6. Quick recall check

Question

1. $s = 30/1500 = 0.02$, so $f_2 = 0.02 \times 50 =$ **1 Hz**.
2. R_2': rotor copper loss (heat), sP_{AG}. $R_2'(1 - s)/s$: power converted to mechanical, $(1 - s)P_{AG}$.
3. Rotor Cu = **500 W**, P_M = **9.5 kW**.
4. $\tau = P_{AG}/\omega_S$ (synchronous). $\tau = P_M/\omega$ (rotor). Both give the same τ.
5. Delta, so $V_1 = V_L =$ **690 V**.